

Calculating the Expected Value Function of a Two-Stage Stochastic Optimization Program with a Quantum Algorithm

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Paper: [arXiv:2402.15029](https://arxiv.org/abs/2402.15029) [quant-ph]

Code: github.com/NatLabRockies/quantum_stochastic_programming

1 Two-Stage Stochastic Unit Commitment

Setting: First-stage gas commitment *before* wind is known; second-stage dispatch *after*.

Objective – minimise total expected cost:

$$\min_x \left[\underbrace{c_g x}_{\text{gas cost}} + \underbrace{\mathbb{E}_\xi \left[\min_{y,r} \{ c_w^\top y + c_r r \} \right]}_{\phi(x) \text{ (expected value function)}} \right]$$

$x \in \{0, 1\}^{n_g}$ commit gas; $y \in \{0, 1\}^{n_y}$ dispatch turbines; $r \geq 0$ recourse; $\xi \sim p(\xi)$ wind scenarios.

Challenge: Computing $\phi(x)$ over 2^{n_y} scenarios is **#P-Hard** classically. Classical Monte Carlo needs $O(1/\varepsilon^2)$ samples for error ε .

Method	Sample complexity	Space
Classical Monte Carlo	$O(1/\varepsilon^2)$	$O(\Omega)$
Quantum DQA + QAE	$O(1/\varepsilon)$	$O(n_y + n_\xi)$

2 Digitized Quantum Annealing (DQA)

DQA encodes **all wind scenarios simultaneously** in a quantum register and anneals the dispatch register to the per-scenario optimal solution.

$$U_{\text{DQA}} = \prod_{t=1}^T e^{-i\gamma_t H_Q} \cdot e^{-i\beta_t H_M}, \quad H_Q = \sum_{\omega} H_{q_\omega} \otimes |\xi_\omega\rangle\langle\xi_\omega|$$

After T steps the state approximates the **per-scenario optimized wavefunction**:

$$|\psi_x^*\rangle \approx \sum_{\omega} \sqrt{p(\omega)} |\xi_\omega\rangle |y_\omega^*\rangle$$

Annealing time $T \sim O(2^{n_y})$ depends only on n_y , not on $|\Omega| = 2^{n_\xi}$.

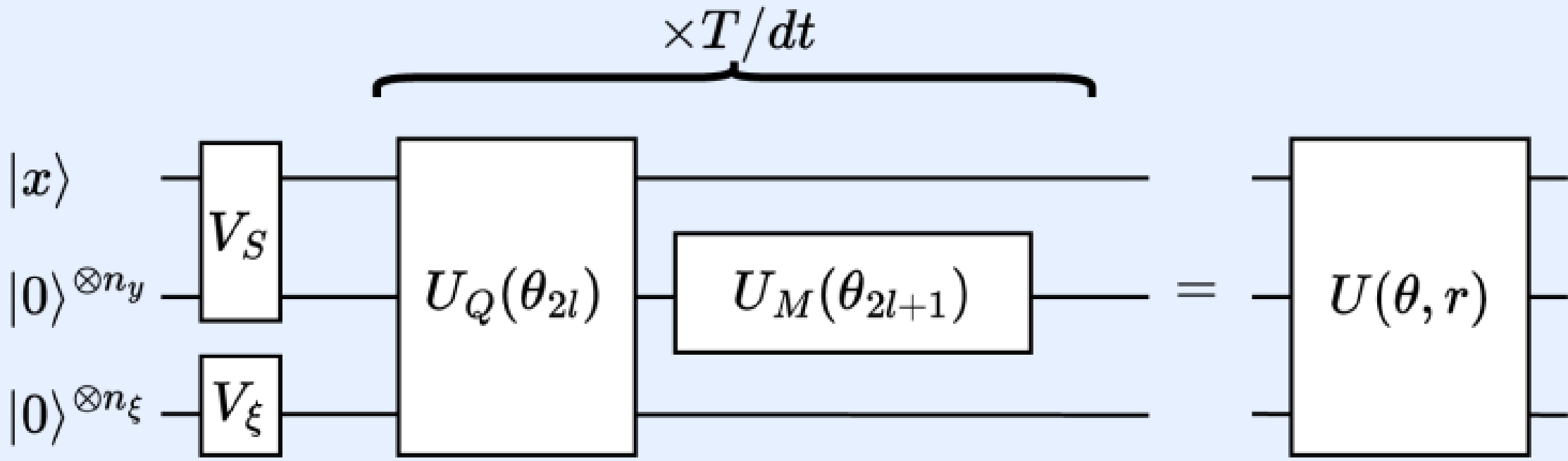


Fig. 1 [1]: DQA circuit. Distribution register $|\xi\rangle$ acts as classical control for U_Q .

3 Quantum Amplitude Estimation (QAE)

QAE wraps DQA in a Quantum Phase Estimation circuit to extract $\phi(x)$ with $O(1/\varepsilon)$ circuit calls.

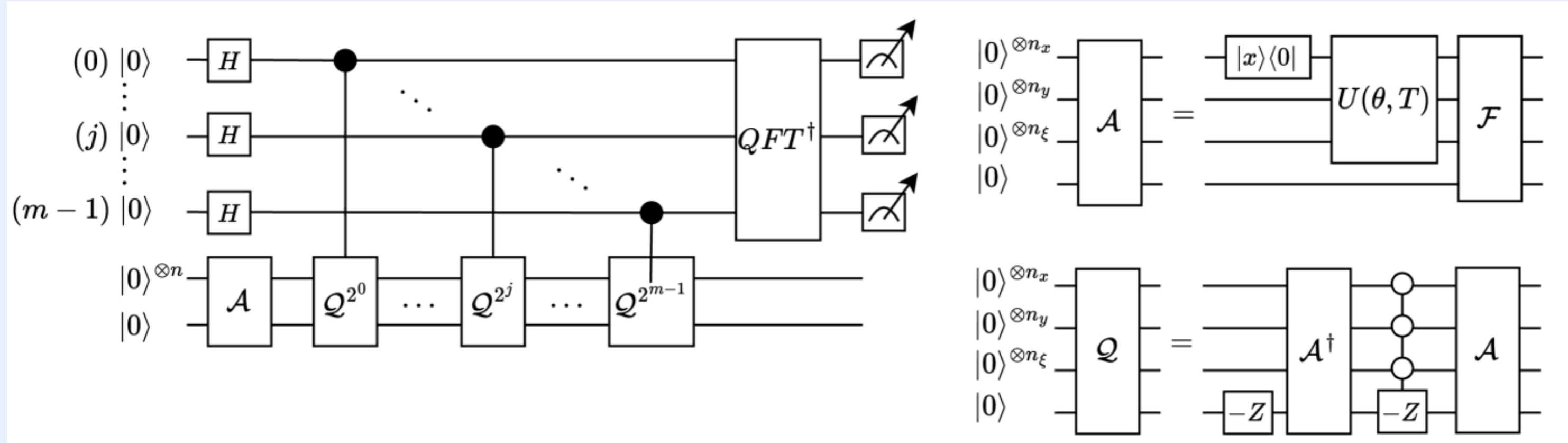


Fig. 2 [1]: QAE circuit. $\mathcal{A} = \mathcal{F} \cdot U(\theta, T)$; $\mathcal{Q}$ is the Grover iterate.

Error bound [1]: two additive sources – DQA residual $\delta \sim O(1/T^2)$ and QAE phase estimation $O(1/M)$:

$$\Pr \left[\left| \tilde{\phi}(x) - \phi(x) - \delta \right| \leq \frac{\pi}{M} + \frac{\pi^2}{M^2} \right] \geq \frac{8}{\pi^2}$$

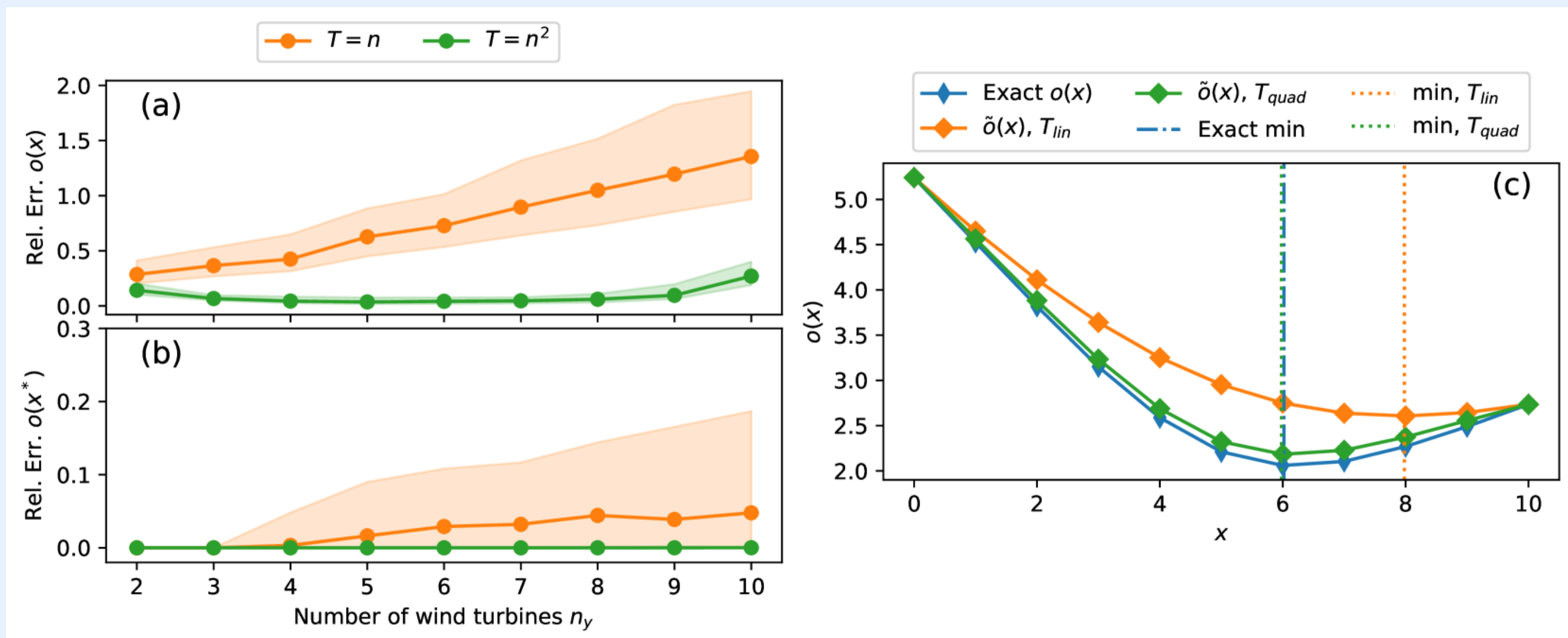


Fig. 3 [1]: DQA convergence. $T = n_y^2$ gives accurate minima.

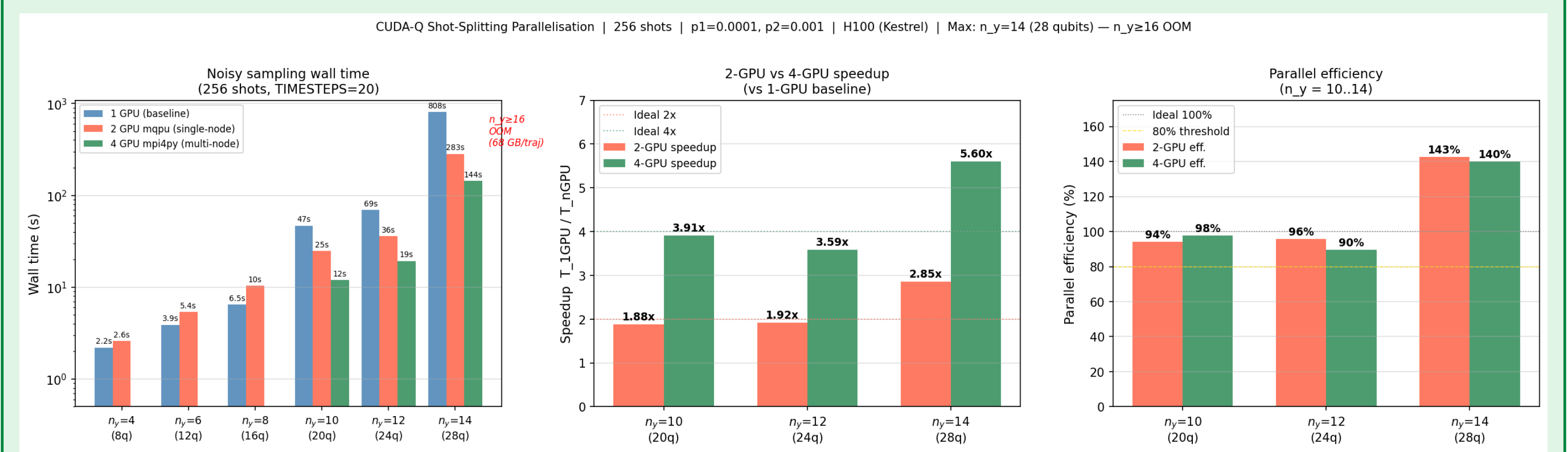
4 Three GPU Parallelisation Strategies

Noisy trajectory simulation holds **one full statevector per shot** on-device ($2^n \times 16$ bytes). Three strategies target this bottleneck on NREL Kestrel H100s (80 GB each).

Strategy Mechanism	Best for
mqpu Each GPU holds a full statevector; shots dispatched via <code>sample_async(qpu_id=i)</code> (CUDA-Q)	$n_y \geq 10$, single node; minimal setup
mpi4py One MPI rank per GPU; independent shots, counts gathered via <code>MPI_Gather</code> (shot-split)	$n_y \geq 10$, multi-node; near-ideal speedup
mgpu Statevector <i>sharded</i> across GPUs via <code>cuStateVec</code> + <code>cuStateVecTL</code> (Noiseless)	$n > 33$ qubits

5 Noisy Benchmark: mqpu vs mpi4py (Kestrel H100s)

Depolarising noise ($p_1 = 10^{-4}$, $p_2 = 10^{-3}$), 256 shots, $T = 20$ DQA steps.

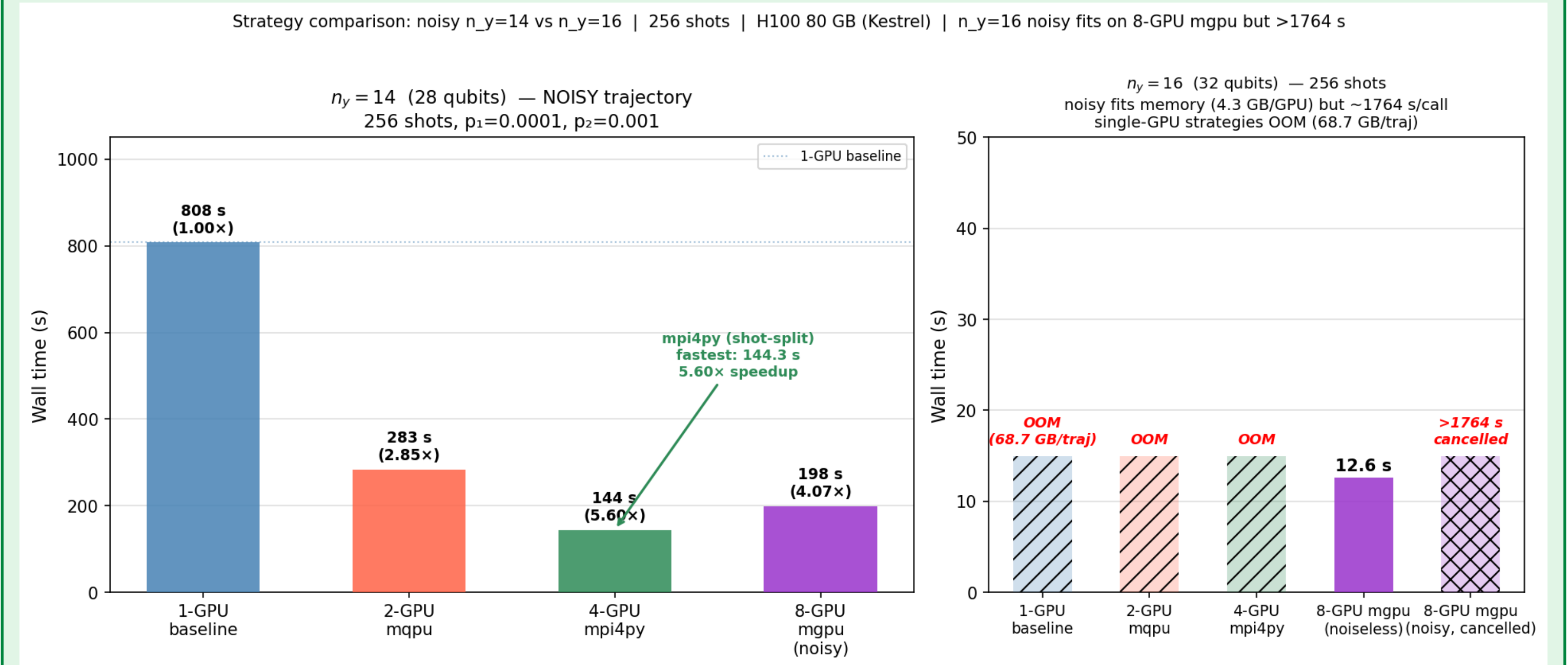


Left: Wall time vs n_y (log scale). mqpu overhead dominates for $n_y \leq 8$; near-2 $\times$ for $n_y \geq 10$.

Centre: Speedup at $n_y = 10$ –14: 4-GPU mpi4py reaches 3.6 $\times$ –5.6 $\times$.

Right: Parallel efficiency > 90% for mpi4py at $n_y = 10$ –12.

6 Strategy Comparison at Scale



Left ($n_y = 14$, noisy): mpi4py 4-GPU = 144 s (5.6 $\times$ vs 1-GPU 808 s). mgpu noisy 8-GPU = 198 s (inter-GPU comms overhead).

Right ($n_y = 16$, 32 q): Single-GPU strategies OOM (68.7 GB / traj). 8-GPU mgpu noiseless: **12.6 s**. Noisy mgpu fits memory (4.3 GB/GPU) but > 1764 s (cancelled).

Key Theoretical Findings

- Polynomial speedup:** QAE estimates $\phi(x)$ with error $O(1/\varepsilon)$ circuit calls vs $O(1/\varepsilon^2)$ for classical Monte Carlo — a provable polynomial advantage assuming efficient distribution preparation.
- Parallel scenario optimisation:** Encoding $p(\xi)$ as a quantum wavefunction and using it as control logic for the phase-separation unitary H_Q lets DQA converge to the optimal dispatch y_ω^* for *all* 2^{n_ξ} scenarios simultaneously in a single quantum circuit.
- Controllable error:** Two independent error sources can each be made arbitrarily small — DQA residual $\delta \sim O(1/T^2)$ via annealing time T , and QPE error $O(\pi/M)$ via register size M . Setting $T = n_y^2$ suffices empirically.
- Complexity separation:** Runtime scales as $O(2^{n_y})$ in annealing time (second-stage turbines) but is *independent* of the number of scenarios $|\Omega| = 2^{n_\xi}$, breaking the classical $O(|\Omega|)$ bottleneck.
- Power grid demonstration:** Algorithm validated on a stochastic unit commitment problem (gas commitment under wind uncertainty), confirming that DQA+QAE recovers the classical expected cost with controllable accuracy.